

SHENGQI ZHANG

Assistant Professor/PhD Supervisor
Eastern Institute of Technology, Ningbo, China
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Education

Ph.D. in Fluid Mechanics	09/2018-07/2022
Peking University (Advisor: Prof. Shiyi Chen)	
B.S. in Theoretical and Applied Mechanics	09/2014-07/2018
Peking University	

Research Interests

Turbulence, heat & mass transfer, multiphase flows, AI-empowered scientific computing

Research Projects

NSFC Young Scientists Fund (National Natural Science Foundation of China), Similarity study of spanwise rotation and wall-normal stratification effects in wall turbulence, ¥300,000, 2024.01-2026.12, PI.

NSFC Joint Fund for Regional Innovation and Development Project – Subproject (National Natural Science Foundation of China), Theoretical Foundations, Algorithms, and Applications of Multiphysics Simulation Based on Artificial Intelligence-Project 5, ¥1,500,000, 2026.01-2029.12, PI.

National Science and Technology Major Project – Subproject (Ministry of Science and Technology of China), Parallel Simulation Algorithm of Macroscale/Microscale Coupling in Forging Process, ¥400,000, 2024.10-2027.09, PI.

Yongjiang Talent Introduction Program (Ningbo Municipal Education Bureau), Numerical investigation on optimal control of rotating wall turbulence, ¥1,000,000, 2025.01-2029.12, PI.

Key R&D Program (Ningbo Municipal Bureau of Science and Technology), Complex flow modelling, fast simulation and software development based on artificial intelligence, ¥1,000,000, 2023.04-2026.04, co-PI.

Top Talent Program (Ningbo Municipal Bureau of Science and Technology), Flow analysis, modelling and simulation software for smart manufacturing, ¥100,000,000, 2023.07-2028.06, Core member.

Industrial R&D Project ([Renewable Energy Company]), Design and numerical simulation of novel fixed-bed reactors, ¥150,000, 2024.10-2025.04, PI.

Honors & Awards

Young Innovative Talent, Science and Technology Department of Zhejiang, 2025

Leading Talent, Ningbo Municipal Human Resources and Social Security Bureau, 2025

Young Innovative Talent, Ningbo Municipal Education Bureau, 2024

Excellent PhD Dissertation, Beijing Municipal Education Commission, 2024

Publications

Feixiong Rao, Shengqi Zhang*, (2026). Unified scaling laws in porous-medium vertical natural convection. *Journal of Fluid Mechanics*, 1028: A17.

Yang Chen, Pingang Wei, Shengqi Zhang, Zhanming Wan, Peng Zhang, Kaixuan Zhang, Peng Wang, Wei Liu*, (2026). Pattern selection between two-dimensional Rayleigh–Bénard convection and electroconvection near electrochemical surfaces. *Journal of Fluid Mechanics*, 1028: A10.

Lei Du, Shengqi Zhang*, Ruili Zhang, Shibin Zhang, (2026). Interpolation-based parametric reduced-order models with dynamic mode decomposition. *Journal of Computational Physics*, 544: 114436.

Zhenze Zhu, Shengqi Zhang*, Lifeng Xiao, (2025). Numerical study of methane dry reforming in an electromagnetic induction heating reactor. *Applied Thermal Engineering*, 280(5): 128514.

Feixiong Rao, Shengqi Zhang*, (2025). Coupled modelling of turbulent forced thermal convection in anisotropic porous media. *Journal of Fluid Mechanics*, 1017: A31.

Lei Du, Shengqi Zhang*, Yongchao Yang, (2025). Interpolation-based parametric reduced-order models via Galerkin projection and dynamic mode decomposition. *Physics of Fluids*, 37(8): 087169.

Feixiong Rao, Shengqi Zhang*, (2025). Heat-fluid-solid coupling model for turbulent forced convection within porous media. *Physical Review Fluids*, 10(6): 064502.

Jin Hu, Shengqi Zhang, Zhenhua Xia*, (2025). An efficient modified enthalpy method for phase-change melting problems with dual time-steps. *International Journal of Heat and Mass Transfer*, 239: 126544.

Kaizhen Shi, Shengqi Zhang*, (2025). Turbulent transport in annular Poiseuille flow with axial rotation. *Physics of Fluids*, 37(3): 035116.

Jun Lai*, Yiming Qi, Shengqi Zhang, Lian-Ping Wang*, (2025). Analytical solutions of layered Poiseuille flows in the diffuse interface model. *Physics of Fluids*, 37(2), 023612.

Feixiong Rao*, Shengqi Zhang, (2025). The in-depth analysis and validation of momentum dispersion modeling based on direct numerical simulation. *Physics of Fluids*, 37(1): 015142.

Qifan Wang, Shengqi Zhang, Taipeng Guo, Wenxuan Sha, Keyang Li, Zhentao Liu*, (2024). Enhancing plate-fin heat exchanger hydraulic thermal performance through air-side fin optimization based on pseudo-3D topology optimization. *Applied Thermal Engineering*, 252: 123642.

Jin Hu, Shengqi Zhang, Zhishun Yang, Lihua Chen, Zhenhua Xia*, (2024). Numerical investigations of melting process with sidewall heating under various control parameters. *International Journal of Heat and Mass Transfer*, 227: 125529.

Jin Hu, Shengqi Zhang, Zhenhua Xia*, (2024). Flow reversal and multiple states in turbulent Rayleigh–Bénard convection with partially isothermal plates. *Journal of Fluid Mechanics*, 987: A9.

Shengqi Zhang*, Chao Sun, (2024). Twin forces: similarity between rotation and stratification effects on wall turbulence. *Journal of Fluid Mechanics*, 979: A45.

Shengqi Zhang, Zhenhua Xia*, Shiyi Chen, (2024). Enhancing large-scale motions and turbulent transport in rotating plane Poiseuille flow. *Journal of Fluid Mechanics*, 979: A43.

Qifan Wang, Shengqi Zhang, Yu Zhang, Jiahong Fu, Zhentao Liu*, (2023). Enhancing performance of nanofluid mini-channel heat sinks through machine learning and multi-objective optimization of operating parameters. *International Journal of Heat and Mass Transfer*, 210: 124204.

Jun Lai, Tao Chen, Shengqi Zhang, Zuoli Xiao, Shiyi Chen, Lian-Ping Wang*, (2022). A systematic study of a droplet breakup process in decaying homogeneous isotropic turbulence using a mesoscopic simulation approach. *Journal of Turbulence*, 23(11-12): 567-614.

Jin Hu, Shengqi Zhang, Zhenhua Xia*, (2022). Flow field and heat transfer properties in turbulent convection with local heating condition at the bottom wall. *Acta Aerodynamica Sinica*, 40(2): 208-214.

Shengqi Zhang, Zhenhua Xia*, Shiyi Chen, (2022). Flow structures in spanwise rotating plane Poiseuille flow based on thermal analogy. *Journal of Fluid Mechanics*, 933: A24.

Shengqi Zhang, Zhenhua Xia*, Shiyi Chen*, (2022). Perturbation analysis of baroclinic torque in low-Mach-number flows. *Journal of Fluid Mechanics*, 930: A4.

Shengqi Zhang, Zhenhua Xia*, (2022). Capturing the Baroclinic Effect in non-Boussinesq Gravity Currents. *Theoretical and Applied Mechanics Letters*, 12(1): 100313.

Shengqi Zhang, Xin Chen, Zhenhua Xia*, Heng-Dong Xi, Quan Zhou, Shiyi Chen*, (2021). Stabilizing/destabilizing the large-scale circulation in turbulent Rayleigh–Bénard convection with sidewall temperature control. *Journal of Fluid Mechanics*, 915: A14.

Shengqi Zhang, Zhenhua Xia*, Shiyi Chen, (2021). A CFD-aided Galerkin Method for Global Linear Instability Analysis. *Communications in Computational Physics*, 29(1): 128-147.

Shengqi Zhang, Zhenhua Xia*, Quan Zhou, Shiyi Chen*, (2020). Controlling flow reversal in two-dimensional Rayleigh–Bénard convection. *Journal of Fluid Mechanics*, 891: R4.

Shengqi Zhang, Zhenhua Xia*, Yipeng Shi, Shiyi Chen*, (2019). A two-dimensional-three-component model for spanwise rotating plane Poiseuille flow. *Journal of Fluid Mechanics*, 880: 478-496.

Liren Li, Yipeng Shi, Shengqi Zhang, Lian-Ping Wang, Zhenhua Xia*, (2019). On the comparison between lattice boltzmann methods and spectral methods for DNS of incompressible turbulent channel flows on small domain size. *Advances in Applied Mathematics and Mechanics*, 11(3): 598-607.

Patents & Software Copyrights

Jun Lai, Shengqi Zhang, (2026). Numerical simulation method and system for two-phase flow and heat transfer. Invention Patent, ZL202411602864.0.

Lei Du, Shengqi Zhang, (2025). DMD parametric reduced-order simulation software V1.0. Software Copyright, 2025SR2160817.

Ruili Zhang, Shengqi Zhang, (2025). Riemannian-metric-based two-dimensional curved mesh generation and adaptation software V1.0. Software Copyright, 2025SR1719524.

Lei Du, Shengqi Zhang, (2025). DMD-Galerkin parametric reduced-order simulation software V1.0. Software Copyright, 2025SR1740896.

Invited Talks

“Validity of buoyancy approximations in low-Mach-number flows”, *China-Japan-Korea Symposium on Thermal and Fluid Engineering*, Jan 2026, Hangzhou China.

“Similarity between stratification and spanwise rotation in shear flows”, *Greater-Southwest Symposium on Applied Mathematics and Mechanics*, Aug 2025, Liangshan, China.

“Similarity between stratification and spanwise rotation in shear flows”, *National Conference on Turbulence and Flow Stability*, Apr 2025, Nanchang, China.

“Similarity between spanwise rotation and vertical stratification in wall turbulence”, *NSFC Key Project Conference*, Jan 2024, Lingshui, China.

“Statistics and parameter reduction of rotating stratified wall turbulence”, *Advanced Summer Program on Turbulence, Noise and CFD Methods*, Aug 2023, Hong Kong, China.

“Similarity between spanwise rotation and vertical stratification in wall turbulence”, *National Conference on Turbulence and Flow Stability*, Apr 2023, Ningbo, China.

“Perturbation analysis of baroclinic torque in low-Mach-number flows”, *NSFC Key Project Conference*, Jan 2023, Ningbo, China.

“Perturbation Analysis and Perturbative Control of Buoyancy-Driven Flows”, *Hong Kong Polytechnic University*, Jan 2023, Hong Kong, China.

“Perturbation analysis of baroclinic torque in low-Mach-number flows”, *Central South University*, Nov 2022, Changsha, China.

Academic Part-time Jobs

Journal reviewer: *Acta Mechanica Sinica*, *Physics of Fluids*, *Journal of Turbulence*, *Scientific Reports*, *Boundary Value Problems*, *Acta Aerodynamica Sinica*, *Chinese Journal of Theoretical and Applied Mechanics*

Ph.D. dissertation reviewer: *China Academic Degrees and Graduate Education Development Center*

Science Popularization Committee member of *Zhejiang Society of Theoretical and Applied Mechanics*